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DEPOLARIZER, DEPOLARIZER ADJUSTMENT METHOD, AND DEPOLARIZER MANUFACTURING METHOD

Abstract

A depolarizer includes: a splitter configured to branch an input light into at least a first light and a second light; a delay applying portion configured to apply a differential group delay with respect to the second light to the first light; a polarization converter configured to convert a polarization state of either the first light or the second light into an orthogonal polarization state; a coupler configured to either multiplex the first light, which has passed through the delay applying portion and the polarization converter, and the second light, or multiplex the first light which has passed through the delay applying portion and the second light which has passed through the polarization converter; and a connecting optical waveguide configured to optically connect the splitter and the coupler.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/043727, filed on Dec. 6, 2023 which claims the benefit of priority of the prior Japanese Patent Application No. 2022-205785, filed on Dec. 22, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a depolarizer, a depolarizer adjustment method, and a depolarizer manufacturing method.

2. Description of the Related Art

[0003] A technology has been disclosed about a depolarizer that lowers the degree of polarization of the light input thereto (refer to Japanese Patent Application Laid-open No. H7-159632 and Japanese Patent Application Laid-open No. 2013-152295). In Japanese Patent Application Laid-open No. H7-159632 and Japanese Patent Application Laid-open No. 2013-152295, in a unit that applies a differential group delay (DGD) to the light, or in a splitter that splits the light, or in a coupler that multiplexes the light; a polarization maintaining optical fiber such as a PANDA fiber is used, or an optical waveguide such as a waveguide core included in a planar lightwave circuit (PLC) is used.

SUMMARY OF THE INVENTION

[0004] In some embodiments, a depolarizer includes: a splitter including an optical waveguide, the splitter being configured to branch an input light into at least a first light and a second light to output the first light and the second light; a delay applying portion including optical waveguides including a plurality of curved optical wavelengths, the delay applying portion being configured to apply a differential group delay with respect to the second light to the first light to output the first light having the differential group delay; a polarization converter configured to convert a polarization state of either the first light or the second light into an orthogonal polarization state to output a post-conversion polarization light; a coupler comprising an optical waveguide, the coupler being configured to either multiplex the first light, which has passed through the delay applying portion and the polarization converter, and the second light, or multiplex the first light which has passed through the delay applying portion and the second light which has passed through the polarization converter to output a multiplexed light; and a connecting optical waveguide configured to optically connect the splitter and the coupler to guide the second light, the splitter, the delay applying portion, the polarization converter, the coupler, and the connecting optical waveguide being integrated on a single optical waveguide substrate.

[0005] In some embodiments, provided is a depolarizer adjustment method of adjusting the depolarizer. The method includes: inputting a light to the depolarizer; receiving a light that is output from the depolarizer; and adjusting, according to an intensity of the received light, branching either a ratio of the splitter or a multiplexing ratio of the coupler.

[0006] In some embodiments, a depolarizer manufacturing method includes: forming a splitter, a

delay applying portion, a polarization converter, a coupler, and a connecting optical waveguide; and implementing the depolarizer adjustment method.

[0007] The above and other objects, features, advantages and technical and industrial significance of this disclosure will be better understood by reading the following detailed description of presently preferred embodiments of the disclosure, when considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram of a depolarizer according to a first embodiment;

[0009] FIG. 2 is a schematic diagram of a depolarizer according to a second embodiment;

[0010] FIG. 3 is a schematic diagram of a depolarizer according to a third embodiment;

[0011] FIG. 4 is a schematic diagram of a depolarizer according to a fourth embodiment;

[0012] FIG. 5A is an A-A cross-sectional view in FIG. 4;

[0013] FIG. 5B is a B-B cross-sectional view in FIG. 4;

[0014] FIG. 5C is a C-C cross-sectional view in FIG. 4;

[0015] FIG. 6 is a schematic diagram of a depolarizer according to a fifth embodiment;

[0016] FIG. 7 is a schematic diagram of a depolarizer according to a sixth embodiment;

[0017] FIG. 8 is a schematic diagram of a depolarizer according to a seventh embodiment;

[0018] FIG. 9 is a schematic diagram of a depolarizer according to an eighth embodiment; and

[0019] FIG. 10 is a schematic diagram of a depolarizer according to a ninth embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0020] Exemplary embodiments are described below with reference to the accompanying drawings. However, the disclosure is not limited by the embodiments described below. In the drawings, identical or corresponding constituent elements are referred to by the same reference numerals, and the same explanation is not repeated. Meanwhile, the drawings are schematic in nature, and it needs to be noted that the relationship among the dimensions of the constituent elements or the proportion of the constituent elements may be different than the actual situation. Among the drawings too, sometimes the relationship among the dimensions and the proportion can be different.

First Embodiment

Configuration of Depolarizer

[0021] FIG. 1 is a schematic diagram of a depolarizer according to a first embodiment. A depolarizer **100** includes a splitter **110**, a delay applying portion **120**, a polarization converter **130**, a coupler **140**, and a connecting optical waveguide **150**. The splitter **110**, the delay applying portion **120**, the polarization converter **130**, the coupler **140**, and the connecting optical waveguide **150** are integrated on an optical waveguide substrate **101**.

[0022] The optical waveguide substrate **101** is a planar lightwave circuit (PLC) and includes end faces **101a** and **101b** positioned opposite to each other. The optical waveguide substrate **101** includes a waveguide core representing an optical waveguide, a cladding that surrounds the waveguide core, and a substrate on the surface of which are formed the optical waveguide and the cladding.

[0023] The substrate is, for example, a silicon substrate or a glass substrate. The cladding is made of a silica based glass material. The waveguide core is made of a silica based glass material having a higher refractive index than the refractive index of the cladding. As a silica based glass material having a high refractive index, for example, it is possible to use silica based glass that contains germanium (GeO.sub.2) or zirconia (ZrO.sub.2) as the dopant for enhancing the refractive index. The relative refractive-index difference of the waveguide core with respect to the cladding is, for example, equal to or greater than 0.8%. If the material constituting the waveguide core is, what is

called, an SiO.sub.2—ZrO.sub.2-based material represented by a silica based glass containing zirconia, the relative refractive-index difference of the waveguide core with respect to the cladding can be enhanced to be equal to or greater than 4.5%, for example.

[0024] The splitter **110** includes an optical waveguide. The splitter **110** includes two input ports **111** and **112**, two output ports **113** and **114**, and a property adjuster **115**. The splitter **110** branches the input light, which is input to the input port **111** or the input port **112**, into a first light and a second light having the same polarization state as the polarization state of the input light; and outputs the first light and the second light from the output ports **113** and **114**, respectively. The intensity ratio between the first light and the second light, that is, the branching ratio of the splitter **110** is desirably equal to 1:1.

[0025] The property adjuster **115** is, for example, a heater formed on the surface of the optical waveguide substrate **101**. The property adjuster **115** is capable of heating some part of the optical waveguide constituting the splitter **110**, and thus changing the refractive index of that part. As a result, the branching ratio of the splitter **110** can be changed on a permanent basis.

[0026] Although there is no particular restriction on the structure of the splitter **110**, for example, the splitter **110** has the structure of a Mach-Zehnder interferometer (MZI).

[0027] The delay applying portion **120** includes optical waveguides. More specifically, the delay applying portion **120** includes: a plurality of curved optical waveguides **121**, **122**, **123**, **124**, **125**, **126**, **127**, and **128**; and linear optical waveguides that are disposed in between the curved optical waveguides. The delay applying portion **120** is connected to the output port **113** of the splitter **110**.

[0028] The coupler **140** includes an optical waveguide. The coupler **140** includes two input ports **141** and **142**, two output ports **143** and **144**, and a property adjuster **145**. The coupler **140** multiplexes the input lights, which are input to the input ports **141** and **142**, without changing their polarization states; and outputs the multiplexed light to the two output ports **143** and **144**. The multiplexing ratio of the coupler **140** is desirably equal to 1:1. Regarding the ratio of the intensity of the light output from the output port **143** and the intensity of the light output from the output port **144**; although there is no particular restriction, the ratio is in the range of 80:20 to 99:1, for example.

[0029] The property adjuster **145** is, for example, a heater formed on the surface of the optical waveguide substrate **101**. The property adjuster **145** is capable of heating some part of the optical waveguide constituting the coupler **140**, and thus changing the refractive index of that part. As a result, the multiplexing ratio of the coupler **140** can be changed on a permanent basis.

[0030] Although there is no particular restriction on the structure of the coupler **140**, for example, the coupler **140** has the structure of a Mach-Zehnder interferometer (MZI).

[0031] On the end face **101b** of the optical waveguide substrate **101**, an optical receiver PD is disposed to be optically connected to the output port **114**. The optical receiver PD is, for example, a photodiode.

[0032] The connecting optical waveguide **150** optically connects the output port **114** of the splitter **110** and the input port **142** of the coupler **140**.

[0033] The polarization converter **130** converts the polarization state of the input light into an orthogonal polarization state that is a polarization state in which a polarization direction is orthogonal to a polarization direction of the input light, and outputs the post-polarization-conversion light. In the first embodiment, the polarization converter **130** is, for example, a filter-type polarization rotating element and is, for example, a half-wave plate. The polarization converter **130** is disposed midway in the connecting optical waveguide **150**. More particularly, the polarization converter **130** is inserted into a slit **101c** that is formed on the optical waveguide substrate **101** while traversing the connecting optical waveguide **150**.

Operations of Depolarizer

[0034] Given below is the explanation of the operations performed when a light Li is input to the depolarizer **100**. The following explanation is given about the case in which the light Li has a TE

polarization that is a polarization direction parallel to the substrate face of the optical waveguide substrate **101**. However, there is no particular restriction on the polarization state of the light Li. [0035] When the light Li is input to the input port **111** of the splitter **110** from the end face **101a** of the optical waveguide substrate **101**; the splitter **110** branches the light Li into a first light Li1 and a second light Li2, and outputs the first light Li1 from the output port **113** and outputs the second Li2 from the output port **114**. The first light Li1 and the second light Li2 have the TE polarization. [0036] The connecting optical waveguide **150** guides the second light Li2. The polarization converter **130**, which is disposed midway in the connecting optical waveguide **150**, converts the polarization state of the second light Li2 into the orthogonal polarization state, and outputs the post-polarization-conversion light to the remaining part of the connecting optical waveguide **150**. More particularly, due to the polarization converter **130**, the polarization state of the second light Li2 is converted from the TE polarization to TM polarization that is orthogonal to the TE polarization. The polarization direction of the TM polarization is perpendicular to the substrate face of the optical waveguide substrate **101**.

[0037] The delay applying portion **120** guides and outputs the first light Li1. At the time of guiding the first light Li1, the delay applying portion **120** applies a differential group delay (DGD) with respect to the second light Li2 to the first light Li1 and outputs the resultant light. As a result, the phase relationship between the first light Li1 and the second light Li2 ceases to exist. In order to eliminate the phase relationship between the first light Li1 and the second light Li2, it is desirable to configure the depolarizer **100** in such a way that the optical path length of the optical waveguide constituting the delay applying portion **120** is longer than the optical path length of the connecting optical waveguide **150** by a length equal to or greater than the coherent length of the light Li. Meanwhile, the first light Li1 that is output from the delay applying portion **120** has TE polarization as the polarization state.

[0038] The coupler **140** multiplexes the lights input to the input ports **141** and **142**, that is, multiplexes the first light Li1, which has passed through the delay applying portion **120** and which has been applied with the differential group delay (DGD), and the second light Li2, which has passed through the polarization converter **130** and which has been subjected to the conversion of the polarization state; and outputs the multiplexed light as a light Lo1 from the output port **143**. The light Lo1 is output from the end face **101b** of the depolarizer **100**. Thus, the light Lo1 output in this manner is a depolarized light.

[0039] The coupler **140** outputs the multiplexed light as a light Lo2 from the output port **144**. The light Lo2 too is a depolarized light. The light Lo2 is received by the optical receiver PD. Then, according to the intensity of the received light Lo2, the optical receiver PD outputs an electric signal. The electric signal can be used in, for example, monitoring the optical loss occurring in the depolarizer **100**.

[0040] In the depolarizer **100** configured in the manner explained above, the splitter **110**, the delay applying portion **120**, the polarization converter **130**, the coupler **140**, and the connecting optical waveguide **150** are integrated on the single optical waveguide substrate **101**. Hence, it becomes possible to eliminate such connection points at which there occurs deviation in the angle between the polarization principal axes. As a result, in the depolarizer **100**, a decline in the properties is held down. Moreover, in the depolarizer **100**, since the delay applying portion **120**, which applies a differential group delay (DGD), is configured with optical waveguides including a plurality of curved optical waveguides; it becomes easier to make the depolarizer **100** compact.

[0041] Moreover, in the depolarizer **100**, the splitter **110** includes the two input ports **111** and **112** and includes the two output ports **113** and **114**. With reference to FIG. 1, the light Li is input from the input port **111**. However, even if the light to be depolarized is input from the input port **112**, the depolarizer **100** still functions as a depolarizer. In that regard, the depolarizer **100** can be used in a such a way that, for example, the input port **111** is connected to a particular light source so as to receive input of the light Li from that light source, and the input port **112** is connected to another

light source so as to receive input of the light from that other light source. In this way, the depolarizer **100** can be used as a shared depolarizer for two different light sources.

[0042] Furthermore, in the depolarizer **100**, the coupler **140** includes the two input ports **141** and **142** and includes the two output ports **143** and **144**. Hence, one of the two output ports **143** and **144** can be used in monitoring the properties of the depolarizer **100**. Particularly, in the depolarizer **100**, since the output port **144** is connected to the optical receiver PD, the optical loss occurring in the depolarizer **100** can be monitored according to the intensity of the received light of the optical receiver PD.

[0043] As a property of the depolarizer **100**, it is desirable that the light Lo1, which is output, has the degree of polarization equal to or smaller than 0.1.

[0044] In the depolarizer **100**, closer the intensity ratio between the first light Li1 and the Li2, which are multiplexed in the coupler **140**, to 1:1, the more excellent are the properties of the depolarizer **100**; and, for example, the degree of polarization of the first light Lo1 can be further reduced. Moreover, in the depolarizer **100**, closer the polarization states of the first light Li1 and the second light Li2, which are multiplexed in the coupler **140**, to the orthogonal state, the more excellent are the properties of the depolarizer **100**.

[0045] Regarding the first light Li1 and the second light Li2 that are multiplexed in the coupler **140**, the reasons for the deviation in the intensity ratio are as follows: the branching ratio of the splitter **110** deviates from 1:1; the optical loss suffered when the first light Li1 is guided is different than the optical loss suffered when the second light Li2 is guided; and the multiplexing ratio of the coupler **140** deviates from 1:1.

[0046] In that regard, for example, in the process for manufacturing the depolarizer **100**, the branching ratio of the splitter **110** can be changed using the property adjuster **115**, and the multiplexing ratio of the coupler **140** can be changed using the property adjuster **145**; so that the intensity ratio between the first light Li1 and the second light Li2, which are multiplexed in the coupler **140**, can be brought closer to 1:1 and the properties of the depolarizer **100** can be improved.

[0047] Meanwhile, regarding the first light Li1 and the second light Li2 that are multiplexed in the coupler **140**, the reason for deviation of the polarization state from the orthogonal state is a phenomenon in which there occurs occasional deviation in the polarization states of the curved optical waveguides **121** to **128** included in the delay applying portion **120** (for example, refer to Japanese Patent No. 5959505 and Japanese Patent No. 7042763).

[0048] In that regard, in the depolarizer **100**, if the sum of the products of the curvature signs and the bending angles of the curved optical waveguides **121** to **128** is equal to zero, it is a desirable condition because of the fact that the deviation in the polarization state is held down. A curvature sign represents the sign of the curvature of a curved optical waveguide. For example, when the guiding direction of the light in a curved optical waveguide is counterclockwise, the curvature sign is positive; and, when the guiding direction is clockwise, the curvature sign is negative. Thus, the curved optical waveguides **121**, **122**, **123**, and **128** have the negative curvature sign, and the curved optical waveguides **124** to **127** have the positive curvature sign. A bending angle represents the angle made by the arc of a curved optical waveguide. In the case of the curved optical waveguide **126**, an angle θ represents the bending angle as illustrated in FIG. 1. The angle θ is substantially equal to 90°.

Second Embodiment

[0049] FIG. 2 is a schematic diagram of a depolarizer according to a second embodiment. As compared to the configuration of the depolarizer **100** according to the first embodiment, a depolarizer **100A** includes an optical waveguide substrate **101A**, which is a PCL, in place of the optical waveguide substrate **101**; includes a splitter **110A** in place of the splitter **110**; and additionally includes a passing-light optical waveguide **151**. Meanwhile, the splitter **110A**, the delay applying portion **120**, the polarization converter **130**, the coupler **140**, the connecting optical

waveguide **150**, and the passing-light optical waveguide **151** are integrated on the optical waveguide substrate **101A**.

[0050] The splitter **110A** includes an optical waveguide. The splitter **110A** includes a single input port **111**; includes three output ports **113**, **114**, and **116**; and includes the property adjuster **115**. The splitter **110A** branches the input light, which is input to the input port **111**, into a first light, a second light, and a third light; and outputs the first light, the second light, and the third light from the three output ports **113**, **114**, and **116**, respectively. The intensity ratio between the first light and the second light, that is, the branching ratio of the splitter **110A** regarding the output ports **113** and **114** is desirably equal to 1:1. Regarding the branching ratio of the splitter **110A** regarding the output ports **113** and **116**; although there is no particular restriction, the branching ratio is in the range of 80:20 to 99:1, for example.

[0051] The passing-light optical waveguide **151** is optically connected to the output port **116** of the splitter **110A**, and extends up to the end face **101b** of the optical waveguide substrate **101A** without involving the coupler **140**. The passing-light optical waveguide **151** is disposed in such a way that the slit **101c**, into which the polarization converter **130** is inserted, traverses the passing-light optical waveguide **151**.

[0052] Given below is the explanation about the operations performed when the light **Li** is input to the depolarizer **100A**. The following explanation is given about the case in which the light **Li** includes TE polarization. However, there is no particular restriction on the polarization state of the light **Li**. When the light **Li** is input from the end face **101a** of the optical waveguide substrate **101A** to the input port **111** of the splitter **110A**, the splitter **110A** branches the light **Li** into the first light **Li1**, the second light **Li2**, and a third light **Li3**; and outputs the first light **Li1**, the second light **Li2**, and the third light **Li3** from the three output ports **113**, **114**, and **116**, respectively.

[0053] Meanwhile, the connecting optical waveguide **150**, the polarization converter **130**, the delay applying portion **120**, and the coupler **140** perform same operations as the operations performed in the depolarizer **100**.

[0054] The passing-light optical waveguide **151** causes the third light **Li3** to pass through the polarization converter **130** and ensures that the light is output from the end face **101b** of the depolarizer **100A** without involving the coupler **140**.

[0055] In an identical manner to the depolarizer **100** according to the first embodiment, in the depolarizer **100A** configured in the manner explained above, a decline in the properties is held down and it becomes easier to make the depolarizer **100A** compact. Moreover, the output port **144** of the coupler **140** can be used in monitoring the properties of the depolarizer **100A**, and the properties of the depolarizer **100A** can be improved using the property adjuster **115** and the property adjuster **145**.

[0056] Moreover, in the depolarizer **100A**, the passing-light optical waveguide **151** causes the third light **Li3** to pass through the polarization converter **130** and ensures that the light is output from the depolarizer **100A** without involving the coupler **140**. In that case, as a result of monitoring the light intensity of the third light **Li3**, it becomes possible to obtain the information regarding the optical loss occurring in the polarization converter **130**. The information regarding the optical loss occurring in the polarization converter **130** can be used to know the difference in the optical loss occurring between the path of the first light **Li1** and the path of the second light **Li2** until the first light **Li1** and the second light **Li2** are input to the coupler **140** in the depolarizer **100A**, and to estimate the properties of the depolarizer **100A**.

Third Embodiment

[0057] FIG. **3** is a schematic diagram of a depolarizer according to a third embodiment. As compared to the configuration of the depolarizer **100** according to the first embodiment, a depolarizer **100B** includes an optical waveguide substrate **101B**, which is a PCL, in place of the optical waveguide substrate **101**; includes a splitter **110B** in place of the splitter **110**; includes a coupler **140B** in place of the coupler **140**; and additionally includes polarizers **161**, **162**, and **163**.

Meanwhile, the splitter **110B**, the delay applying portion **120**, the polarization converter **130**, the coupler **140B**, the connecting optical waveguide **150**, and the polarizers **161**, **162**, and **163** are integrated on the optical waveguide substrate **101B**.

[0058] The splitter **110B** includes an optical waveguide. The splitter **110B** includes a single input port **111**; two output ports **113** and **114**; and the property adjuster **115**. The splitter **110B** branches the input light, which is input to the input port **111**, into the first light and the second light, and outputs the first light and the second light from the output ports **113** and **114**, respectively. The intensity ratio between the first light and the second light, that is, the branching ratio of the splitter **110B** regarding the output ports **113** and **114** is desirably equal to 1:1.

[0059] The coupler **140B** includes an optical waveguide. The coupler **140B** includes two input ports **141** and **142**; a single output port **143**; and the property adjuster **145**. The coupler **140B** multiplexes the input lights that are input to the input ports **141** and **142**, and outputs the multiplexed light from the single output port **143**. The branching ratio of the coupler **140** is desirably equal to 1:1.

[0060] The polarizer **161** is disposed on an input side of the splitter **110B**. For example, the polarizer **161** is disposed in between the end face **101a** of the optical waveguide substrate **101B** and the input port **111** of the splitter **110B**. More particularly, the polarizer **161** is inserted into a slit **101Bd** that is formed on the optical waveguide substrate **101B**. The polarizer **161** selectively allows passage of the light that has the TE polarization. Thus, the polarizer **161** represents an example of a polarizer that selectively allows passage of the light that has the same linear polarization as the linear polarization of the light input to the splitter **110B**.

[0061] The polarizer **162** is disposed on an output side of the delay applying portion **120** and on an input side of the coupler **140B**. More particularly, the polarizer **162** is inserted into a slit **101B2** that is formed on the optical waveguide substrate **101B**. The polarizer **162** selectively allows passage of the light that has the TE polarization. Thus, the polarizer **162** represents an example of a polarizer that selectively allows passage of the light that has the same linear polarization as the linear polarization of the light input to the splitter **110B**.

[0062] The polarizer **163** is disposed on an output side of the polarization converter **130** and on an input side of the coupler **140B**. More particularly, the polarizer **163** is inserted into a slit **101Be**. The polarizer **163** selectively allows passage of the light having the TM polarization. Thus, the polarizer **163** represents an example of a polarizer that selectively allows passage of the light that has the linear polarization orthogonal to the linear polarization of the light input to the splitter **110B**.

[0063] Given below is the explanation of the operations performed when the light **Li** is input to the depolarizer **100B**. The light **Li** includes the TE polarization representing linear polarization.

[0064] When input from the end face **101a** of the optical waveguide substrate **101B**, the light **Li** passes through the polarizer **161** and enters the input port **111** of the splitter **110B**. The splitter **110B** branches the light **Li** into the first light **Li1** and the second light **Li2**, and outputs the first light **Li1** and the second light **Li2** from the output ports **113** and **114**, respectively.

[0065] When the polarization state of the input light **Li** deviates from the TE polarization, the light **Li** happens to include the component of the TE polarization and the component of the TM polarization. In that case, when the light **Li** is input without modification to the splitter **110B**, there are times when the depolarizer properties decline as compared to the original properties. In contrast, in the depolarizer **100B**, the light **Li** passes through the polarizer **161** and enters the splitter **110B** in the state of having significant attenuation in the component of the TM polarization. Hence, a decline in the properties of the depolarizer **100B** is held down.

[0066] The connecting optical waveguide **150**, the polarization converter **130**, and the delay applying portion **120** perform same operations as the operations performed in the depolarizer **100**.

[0067] The coupler **140B** multiplexes the lights input to the input ports **141** and **142**, that is, multiplexes the first light **Li1**, which has passed through the delay applying portion **120** and which

has been applied with the differential group delay (DGD), and the second light Li2, which has passed through the polarization converter **130** and which has been subjected to the conversion of the polarization state; and outputs the multiplexed light as the light Lo1 from the output port **143**. The light Lo1 is output from the end face **101b** of the depolarizer **100B**. Thus, the light Lo1 output in this manner is output as a result of depolarization of the light Li.

[0068] When the polarization state of the first light Li1 deviates from the TE polarization as a result of being guided by the delay applying portion **120**, the first light Li1 happens to include the component of the TE polarization and the component of the TM polarization. In that case, when the first light Li1 is input without modification to the coupler **140B** and is multiplexed with the second light Li2, there are times when the depolarizer properties decline as compared to the original properties. In contrast, in the depolarizer **100B**, the first light Li1 passes through the polarizer **162** and enters the coupler **140B** in the state of having significant attenuation in the component of the TM polarization. Hence, a decline in the properties of the depolarizer **100B** is held down.

[0069] Moreover, when the polarization state of the second light Li2, which has already passed through the polarization converter **130**, deviates from the TM polarization, the first light Li1 happens to include the component of the TM polarization and the component of the TE polarization. In that case, if the second light Li2 is input without modification to the coupler **140B** and gets multiplexed with the first light Li1, there are times when the depolarizer properties decline as compared to the original properties. In contrast, in the depolarizer **100B**, the second light Li2 passes through the polarizer **163** and enters the coupler **140B** in the state of having significant attenuation in the component of the TE polarization. Hence, a decline in the properties of the depolarizer **100B** is held down.

[0070] In an identical manner to the depolarizer **100** according to the first embodiment, in the depolarizer **100B** configured in the manner explained above, a decline in the properties is held down and it becomes easier to make the depolarizer **100B** compact. Moreover, the properties of the depolarizer **100B** can be improved using the property adjuster **115** and the property adjuster **145**.

[0071] Furthermore, in the depolarizer **100B**, as explained above, a decline in the properties of the depolarizer **100B** is held down using the polarizers **161**, **162**, and **163**.

[0072] Meanwhile, the depolarizer **100B** according to the third embodiment includes the polarizers **161**, **162**, and **163**. However, as long as at least one of the polarizers **161**, **162**, and **163** is included; it becomes possible to hold down a decline in the properties of the depolarizer **100B**.

[0073] Moreover, in the depolarizer **100B** according to the third embodiment, the input light Li has the TE polarization. Alternatively, if the input light Li has the TM polarization, a depolarizer can be configured in such a way that the polarizers **161** and **162** selectively allow passage of the light having the TM polarization and that the polarizer **163** selectively allows passage of the light having the TE polarization. As a result, a decline in the properties of the depolarizer can be held down in an identical manner.

Fourth Embodiment

[0074] FIG. **4** is a schematic diagram of a depolarizer according to a fourth embodiment. As compared to the configuration of the depolarizer **100** according to the first embodiment, a depolarizer **100C** includes an optical waveguide substrate **101C**, which is a PCL, in place of the optical waveguide substrate **101**; and includes a delay applying portion **120C** in place of the delay applying portion **120**. Meanwhile, the splitter **110**, the delay applying portion **120C**, the polarization converter **130**, the coupler **140**, and the connecting optical waveguide **150** are integrated on the optical waveguide substrate **101C**.

[0075] The optical waveguides constituting the delay applying portion **120C** include a first delay optical waveguide **120C1**, a second delay optical waveguide **120C2**, and a third delay optical waveguide **120C3**.

[0076] FIGS. **5A** to **5C** are an A-A cross-sectional view, a B-B cross-sectional view, and a C-C cross-sectional view, respectively, in FIG. **4**. More particularly, FIG. **5A** is an A-A cross-sectional

view, FIG. 5B is a B-B cross-sectional view, and FIG. 5C is a C-C cross-sectional view. In FIGS. 5A to 5C, a substrate **101f**, a cladding **101g**, the first delay optical waveguide **120C1**, the second delay optical waveguide **120C2**, and the third delay optical waveguide **120C3** of the optical waveguide substrate **101C** are illustrated.

[0077] As illustrated in FIGS. 5A and 5C, the waveguide core width representing the width of the second delay optical waveguide **120C2** is greater than the waveguide core width representing the width of the first delay optical waveguide **120C1**. As a result, with respect to a light having the TE polarization that has a polarization direction indicated by an arrow **Ar**, the effective refractive index of the second delay optical waveguide **120C2** is higher than the effective refractive index of the first delay optical waveguide **120C1**. For example, the width of the second delay optical waveguide **120C2** is 10 times greater than the width of the first delay optical waveguide **120C1**, but can alternatively be greater in the range of twice to 15 times.

[0078] As illustrated in FIGS. 4 and 5B, the third delay optical waveguide **120C3** is connected to the first delay optical waveguide **120C1** and the second delay optical waveguide **120C2**. The third delay optical waveguide **120C3** has a continuously-changing optical core width. More particularly, from the first delay optical waveguide **120C1** toward the second delay optical waveguide **120C2**, the optical core width of the third delay optical waveguide **120C3** continuously increases starting from the optical core width of the first delay optical waveguide **120C1** up to the optical core width of the second delay optical waveguide **120C2**.

[0079] The splitter **110**, the polarization converter **130**, the coupler **140**, and the connecting optical waveguide **150** have the same configuration as the configuration in the depolarizer **100** and perform same operations as the operations performed in the depolarizer **100**.

[0080] In an identical manner to the depolarizer **100** according to the first embodiment, in the depolarizer **100C** configured in the manner explained above, a decline in the properties is held down; it becomes easier to make the depolarizer **100A** compact; the output port **144** of the coupler **140** can be used in monitoring the properties of the depolarizer **100C**; and the properties of the depolarizer **100C** can be improved using the property adjuster **115** and the property adjuster **145**.

[0081] Moreover, in the depolarizer **100C**, when the polarization state of the first light **Li1** that is input to the delay applying portion **120C** is TE polarization; a greater differential group delay (DGD) can be applied to the first light **Li1** in the second delay optical waveguide **120C2** that has a higher effective refractive index to the first light **Li1**. As a result, as compared to a delay applying portion that, across the longitudinal direction of the waveguide core width of the optical waveguide constituting that delay applying portion, has the same width as the first delay optical waveguide **120C1**; the delay applying portion **120C** becomes able to apply a greater differential group delay (DGD) over the same length. In other words, in the case of applying the same differential group delay (DGD), it serves the purpose even when the delay applying portion **120C** is shorter, thereby enabling achieving a more compact depolarizer **100C**.

[0082] Moreover, the depolarizer **100C** includes the third delay optical waveguide **120C3**, thereby making it possible to hold down an increase in the connection loss occurring between the first delay optical waveguide **120C1** and the second delay optical waveguide **120C2**.

[0083] In the depolarizer **100C**, the width of the second delay optical waveguide **120C2** is greater than the width of the first delay optical waveguide **120C1**. Hence, with respect to the light having the TE polarization, the effective refractive index of the second delay optical waveguide **120C2** is set to be greater than the effective refractive index of the first delay optical waveguide **120C1**. On the other hand, when the light that is input to the optical waveguide constituting a delay applying portion has the polarization state of TM polarization, by keeping a height of the second delay optical waveguide to be greater than the first delay optical waveguide; with respect to the input light, the effective refractive index of the second delay optical waveguide can be set to be greater than the effective refractive index of the first delay optical waveguide, the height of the second delay optical waveguide being a vertical height from the substrate face of the optical waveguide

substrate. Moreover, when the light that is input to the optical waveguide constituting a delay applying portion has the polarization state including TM polarization and TE polarization, by keeping the width and the height of the second delay optical waveguide to be greater than the width and the height of the first delay optical waveguide; with respect to the input light, the effective refractive index of the second delay optical waveguide can be set to be greater than the effective refractive index of the first delay optical waveguide.

[0084] Meanwhile, in the delay applying portion **120C** too, it is desirable that the sum of the products of the curvature signs and the bending angles of a plurality of curved optical waveguides included in the delay applying portion **120C** is equal to zero.

Fifth Embodiment

[0085] FIG. **6** is a schematic diagram of a depolarizer according to a fifth embodiment. A depolarizer **100D** includes a plurality of splitters **110-1**, **110-2**, . . . , and **110-n**; a plurality of delay applying portions **120-1**, **120-2**, . . . , and **120-n**; a plurality of polarization converter **130-1**, **130-2**, . . . , and **130-n**; a plurality of couplers **140-1**, **140-2**, . . . , and **140-n**; and a plurality of connecting optical waveguides **150-1**, **150-2**, . . . , and **150-n**. Herein, for example, “n” is an integer equal to greater than “3”. The splitters **110-1** to **110-n**, the delay applying portions **120-1** to **120-n**, the polarization converters **130-1** to **130-n**, the couplers **140-1** to **140-n**, and the connecting optical waveguides **150-1** to **150-n** are integrated on an optical waveguide substrate **101D** representing a PLC.

[0086] The splitters **110-1** to **110-n** have an identical configuration and an identical function to the splitter **110**. The delay applying portions **120-1** to **120-n** have an identical configuration and an identical function to the delay applying portion **120**. The polarization converters **130-1** to **130-n** have an identical configuration and identical function to the polarization converter **130**. The couplers **140-1** to **140-n** have an identical configuration and an identical function to the coupler **140**. The connecting optical waveguides **150-1** to **150-n** have an identical configuration and an identical function to the coupler **140**.

[0087] The optical waveguides constituting the delay applying portions **120-1** to **120-n** are arranged parallel to each other in a curved manner on the optical waveguide substrate **101D**.

[0088] The polarization converters **130-1** to **130-n** are inserted in a slit **101Dc** that is formed on the optical waveguide substrate **101D** while traversing the connecting optical waveguides **150-1** to **150-n**.

[0089] The splitter **110-1**, the delay applying portion **120-1**, the polarization converter **130-1**, the coupler **140-1**, and the connecting optical waveguide **150-1** are optically connected to each other, and constitute a single unit depolarizer. In an identical manner, each of the remaining splitters, each of the remaining delay applying portions, each of the remaining polarization converters, each of the remaining couplers, and each of the remaining connecting optical waveguides are optically connected to constitute a single unit polarizer. Thus, n number of unit polarizers are integrated on the depolarizer **100D**.

[0090] In the depolarizer **100D**, one of first lights **Li10**, **Li20**, . . . , and **Lin0** is input to each of the n number of unit depolarizers. Then, each unit depolarizer outputs one of lights **Lo10**, **Lo20**, . . . , and **Lon0** that are obtained as a result of depolarization of the first lights **Li10** to **Lin0**.

[0091] In the depolarizer **100D** configured in the manner explained above, each unit depolarizer enables achieving identical effects to the effects achieved in the depolarizer **100** according to the first embodiment. Moreover, as a result of including n number of unit depolarizers, the depolarizer **100D** becomes able to depolarize n number of lights in a simultaneous manner. Moreover, if each unit depolarizer is configured as a depolarizer having the properties suitable for a different wavelength band, the depolarizer **100D** gets configured to have a wide operating wavelength band. Furthermore, since the optical waveguides constituting the delay applying portions **120-1** to **120-n** are arranged parallel to each other in a curved manner, the depolarizer **100D** can be configured in a more compact manner despite having n number of unit depolarizers integrated therein.

Sixth Embodiment

[0092] FIG. 7 is a schematic diagram of a depolarizer according to a sixth embodiment. As compared to the configuration of the depolarizer **100D** according to the fifth embodiment, a depolarizer **100E** includes an optical waveguide substrate **101E**, which is a PCL, in place of the optical waveguide substrate **101D**; and includes the polarization converter **130** in place of the polarization converters **130-1**, **130-2**, . . . , and **130-n**.

[0093] The polarization converter **130** is disposed midway in the optical waveguides constituting the delay applying portions **120-1** to **120-n**. More particularly, the polarization converter **130** is inserted in a slit **101Ec** that is formed on the optical waveguide substrate **101E** while traversing the optical waveguides constituting the delay applying portions **120-1** to **120-n**. As a result, the optical waveguides constituting the delay applying portions **120-1** to **120-n** are all optically connected to the shared polarization converter **130**.

[0094] The polarization converter **130** converts the polarization state of the first lights, which are guided by the delay applying portions **120-1** to **120-n**, into an orthogonal polarization state, and outputs the post-polarization-conversion lights to the remaining parts of the delay applying portions **120-1** to **120-n**.

[0095] On the other hand, the second lights that are guided by the connecting optical waveguides **150-1** to **150-n** are guided without being subjected to conversion of their polarization state. As a result, each of the couplers **140-1** to **140-n** multiplexes a first light and a second light having mutually orthogonal polarization states.

[0096] In the depolarizer **100E** configured in the manner explained above, it is possible to achieve identical effects to the effects achieved in the depolarizer **100E** according to the fifth embodiment. Moreover, in the depolarizer **100E**, since the polarization converter **130** is shared, a more compact depolarizer **100E** can be achieved as compared to the depolarizer **100D**.

Seventh Embodiment

[0097] FIG. 8 is a schematic diagram of a depolarizer according to a seventh embodiment. As compared to the configuration of the depolarizer **100D** according to the sixth embodiment, a depolarizer **100F** includes an optical waveguide substrate **101F**, which is a PCL, in place of the optical waveguide substrate **101E**, and a slit **101Ff** is formed on the optical waveguide substrate **101F**.

[0098] The slit **101Ff** is formed on the optical waveguide substrate **101F** while traversing the connecting optical waveguides **150-1** to **150-n**. The slit **101Ff** represents an example of a second slit. The slit **101Ec** represents an example of a first slit.

[0099] The depolarizer **100F** configured in the manner explained above enables achieving identical effects to the effects achieved in the depolarizer **100E** according to the sixth embodiment. In the depolarizer **100F**, the slit **101Ec** is formed to traverse the optical waveguides constituting the delay applying portions **120-1** to **120-n**, and the slit **101Ff** is formed to traverse the connecting optical waveguides **150-1** to **150-n**. As a result, between the paths of the lights guided by the delay applying portions **120-1** to **120-n** and the paths of the lights guided by the connecting optical waveguides **150-1** to **150-n**, the optical loss occurring due to the slits either becomes equal to each other or has only a small difference therebetween. Hence, a decline in the depolarizer properties of each unit depolarizer is held down.

[0100] Meanwhile, the effects achieved in the configuration in which a slit is formed to traverse the optical waveguides constituting the delay applying portions and in which a slit is formed to traverse the connecting optical waveguides can also be achieved when the configuration is implemented in a depolarizer including only a single unit depolarizer, such as in the depolarizer **100** according to the first embodiment.

[0101] In the depolarizer **100F**, although the polarization converter is inserted in the first slit, it can be alternatively inserted in the second slit.

Eighth Embodiment

[0102] FIG. 9 is a schematic diagram of a depolarizer according to an eighth embodiment. As compared to the configuration of the depolarizer **100D** according to the fifth embodiment, a depolarizer **100G** includes an optical waveguide substrate **101G**, which is a PCL, in place of the optical waveguide substrate **101D**; includes splitters **110G-1** to **110G-n** in place of the splitters **110-1** to **110-n**; includes couplers **140G-1** to **140G-n** in place of the couplers **140-1** to **140-n**; and additionally includes optical switches **171** and **172**. The splitters **110G-1** to **110G-n**, the delay applying portions **120-1** to **120-n**, the polarization converters **130-1** to **130-n**, the couplers **140G-1** to **140G-n**, the connecting optical waveguides **150-1** to **150-n**, and the optical switches **171** and **172** are integrated on the optical waveguide substrate **101G**.

[0103] Except for the fact that only a single input port is present, the splitters **110G-1** to **110G-n** have an identical configuration and an identical functional to the splitter **110**. Moreover, except for the fact that only a single output port is present, the couplers **140G-1** to **140G-n** have an identical configuration and an identical function to the coupler **140**.

[0104] In an identical manner to the depolarizer **100D**, the depolarizer **100G** includes n number of unit depolarizers.

[0105] The optical switch **171** is disposed on an input side of the splitters **110G-1** to **110G-n**. The optical switch **171** is switch-controlled so as to output the light that is input to the optical switch **171** to one of the splitters **110G-1** to **110G-n**. The optical switch **171** represents an example of an input-side optical switch.

[0106] The optical switch **172** is disposed on an output side of the couplers **140G-1** to **140G-n**. The optical switch **172** is switch-controlled so as to output the light that is input to the optical switch **172** from one of the couplers **140G-1** to **140G-n**. The optical switch **172** represents an example of an output-side optical switch.

[0107] The optical switches **171** and **172** includes, for example, a multistage Mach-Zehnder interferometer (MZI) using the optical waveguide.

[0108] Given below is the explanation of the operations performed in the depolarizer **100G**. When the light **Li** is input to the depolarizer **100G**, the optical switch **171** outputs the light **Li**, which is input from the end face **101a**, to one of the splitters **110G-1** to **110G-n** that is selected according to the switch control. The unit depolarizer that includes the splitter to which the light **Li** is input depolarizes the input light **Li**, and outputs the depolarized light **Lo1** from the coupler included therein to the optical switch **172**. Then, the optical switch **172** outputs the light **Lo1** from the end face **101b**.

[0109] The depolarizer **100G** configured in the manner explained above enables achieving identical effects to the effects achieved in the depolarizer **100D** according to the fifth embodiment.

Moreover, when each unit depolarizer is configured as a depolarizer having the properties suitable for a different wavelength band and is connected to a single broadband light source (for example, a wavelength-tunable light source); the depolarizer **100G** becomes able to depolarize, without having to repeatedly get linked to the broadband light source, the light that is output from the broadband light source and that has a wide wavelength band.

Ninth Embodiment

[0110] FIG. 10 is a schematic diagram of a depolarizer according to a ninth embodiment. As compared to the configuration of the depolarizer **100G** according to the eighth embodiment, a depolarizer **100H** includes an optical waveguide substrate **101H**, which is a PCL, in place of the optical waveguide substrate **101G**; and does not include the optical switch **172**. The splitters **110G-1** to **110G-n**, the delay applying portions **120-1** to **120-n**, the polarization converters **130-1** to **130-n**, the couplers **140G-1** to **140G-n**, the connecting optical waveguides **150-1** to **150-n**, and the optical switch **171** are integrated on the optical waveguide substrate **101H**.

[0111] In the depolarizer **100H**, the input light **Li** is depolarized in one of the n number of unit depolarizers, and the depolarized light is output from one of the couplers **140G-1** to **140G-n** toward the end face **101b**.

[0112] The depolarizer **100H** configured in the manner explained above enables achieving identical effects to the effects achieved in the depolarizer **100D** according to the fifth embodiment. Moreover, the depolarizer **100H** can change the output destination according to the unit depolarizer used in depolarization. In that case, for example, if each unit polarizer is designed for the properties suitable for a different wavelength band, the output destination can be changed according to the wavelength of the light to be depolarized.

Depolarizer Adjustment Method and Depolarizer Manufacturing Method

[0113] Given below is the explanation about a depolarizer adjustment method of adjusting the properties of the depolarizers according to the embodiments described above. The depolarizer adjustment method includes, for example: inputting a light to a depolarizer; receiving a light output from the depolarizer; and adjusting either a branching ratio of a splitter or a multiplexing ratio of a coupler according to an intensity of a light received by an optical receiver. According to the depolarization adjustment method, even after the waveguide structure of the depolarizer has been formed, it becomes possible to adjust the properties of the depolarizer.

[0114] A depolarizer manufacturing method of manufacturing the depolarizers according to the embodiments described above includes, for example: forming a splitter, a delay applying portion, a polarization converter, a camera, and a connecting optical waveguide; and implementing the depolarization adjustment method explained above. According to the depolarizer manufacturing method, even after the waveguide structure of the depolarizer has been formed, it becomes possible to manufacture the depolarizer in the adjusted form.

[0115] In the embodiments described above, although the branching ratio of the splitter as well as the multiplexing ratio of the coupler can be changed, it is alternatively possible to keep only one of the two ratios variable. Moreover, at least either the splitter or the coupler can include two input ports and two output ports.

[0116] In the embodiments described above, a polarization beam combiner (PBC) can be used as the coupler. When a PBC is used as the coupler, two lights having the polarization states in which linear polarizations are orthogonal to each other can be multiplexed at a low loss. Moreover, when a PBC is used as the coupler, multiplexing can be performed while holding down a decline in the degree of polarization of the two lights at the time of multiplexing. As a result, it becomes possible to hold down a decline in the depolarization properties of the depolarizer.

[0117] In the embodiments described above, the optical waveguide substrate is a PLC including the optical waveguide made of a glass material. However, the constituent material of the optical waveguide substrate is not limited to a glass material. For example, silicon or silicon nitride can be used as the constituent material of the optical waveguide substrate, or a semiconductor material such as an indium-phosphide-based semiconductor can be used.

[0118] Meanwhile, although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited. Thus, a configuration obtained by appropriately combining the constituent elements explained above is also included in the disclosure. For example, the configuration of the delay applying portion **120C** in the depolarizer **100C** according to the fourth embodiment can be implemented in the depolarizers according to the other embodiments. Moreover, the appended claims are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

[0119] According to the disclosure, it becomes possible to obtain a depolarizer in which compactness is achieved with ease and a decline in the properties is held down.

[0120] Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A depolarizer comprising: a splitter comprising an optical waveguide, the splitter being configured to branch an input light into at least a first light and a second light to output the first light and the second light; a delay applying portion comprising optical waveguides including a plurality of curved optical wavelengths, the delay applying portion being configured to apply a differential group delay with respect to the second light to the first light to output the first light having the differential group delay; a polarization converter configured to convert a polarization state of either the first light or the second light into an orthogonal polarization state to output a post-conversion polarization light; a coupler comprising an optical waveguide, the coupler being configured to either multiplex the first light, which has passed through the delay applying portion and the polarization converter, and the second light, or multiplex the first light which has passed through the delay applying portion and the second light which has passed through the polarization converter to output a multiplexed light; and a connecting optical waveguide configured to optically connect the splitter and the coupler to guide the second light, the splitter, the delay applying portion, the polarization converter, the coupler, and the connecting optical waveguide being integrated on a single optical waveguide substrate.
2. The depolarizer according to claim 1, wherein at least either the splitter or the coupler includes two input ports and two output ports.
3. The depolarizer according to claim 1, wherein the coupler includes two input ports and two output ports, and one of the two output ports is connected to an optical receiver.
4. The depolarizer according to claim 1, wherein the depolarizer is configured such that at least either a branching ratio of the splitter or a multiplexing ratio of the coupler is variable.
5. The depolarizer according to claim 1, wherein the splitter includes three output ports, and is configured to branch the input light into the first light, the second light, and a third light to output the first light, the second light, and the third light, and the depolarizer further includes a passing-light optical waveguide configured to cause the third light to pass through the polarization converter and output the third light from the depolarizer without involving the coupler.
6. The depolarizer according to claim 1, wherein the polarization converter is disposed midway in an optical waveguide constituting the delay applying portion, and is configured to convert the polarization state of the first light into the orthogonal polarization state to output the first light with the orthogonal polarization state.
7. The depolarizer according to claim 1, wherein the optical waveguide substrate includes a first slit configured to traverse an optical waveguide constituting the delay applying portion, and a second slit configured to traverse the connecting optical waveguide, and the polarization converter is a polarization rotating element that is inserted into either the first slit or the second slit.
8. The depolarizer according to claim 1, wherein the splitter is configured to input a light having a first linear polarization, and the depolarizer further comprises a polarizer that is disposed at least on an input side of the splitter, or on an output side of the delay applying portion and an input side of the coupler, or on an output side of the polarization converter and an input side of the coupler, the polarizer being configured to selectively allow passage of a light having either the first linear polarization or a second linear polarization orthogonal to the first linear polarization.
9. The depolarizer according to claim 1, wherein the coupler is a polarization beam combiner.
10. The depolarizer according to claim 1, wherein the depolarizer is configured such that a sum of products of curvature signs and bending angles of the plurality of curved optical waveguides is equal to zero.
11. The depolarizer according to claim 1, wherein the optical waveguides constituting the delay applying portion include a first delay optical waveguide and a second delay optical waveguide, an effective refractive index of the second delay optical waveguide to an input light being higher than

an effective refractive index of the first delay optical waveguide to an input light.

12. The depolarizer according to claim 11, wherein a waveguide core width of the second delay optical waveguide is greater than a waveguide core width of the first delay optical waveguide.

13. The depolarizer according to claim 12, wherein the optical waveguides constituting the delay applying portion include a third delay optical waveguide that connects the first delay optical waveguide and the second delay optical waveguide and that has a continuously-changing optical core width.

14. The depolarizer according to claim 1, wherein the depolarizer comprises a plurality of sets of the splitter, the delay applying portion, the polarization converter, and the coupler that are optically connected, and optical waveguides constituting a plurality of delay applying portions are arranged parallel to each other in a curved manner.

15. The depolarizer according to claim 1, wherein the depolarizer comprises a plurality of sets of the splitter, the delay applying portion, and the coupler that are optically connected, each set is optically connected in common to the polarization converter, and optical waveguides constituting a plurality of delay applying portions are arranged parallel to each other in a curved manner.

16. The depolarizer according to claim 14, further comprising an input-side optical switch configured to output a light that is input to the input-side optical switch to one of a plurality of splitters.

17. The depolarizer according to claim 16, further comprising an output-side optical switch configured to output a light that is input to the output-side optical switch from one of a plurality of couplers.

18. A depolarizer adjustment method of adjusting the depolarizer according to claim 1, the method comprising: inputting a light to the depolarizer; receiving a light that is output from the depolarizer; and adjusting, according to an intensity of the received light, branching either a ratio of the splitter or a multiplexing ratio of the coupler.

19. A depolarizer manufacturing method comprising: forming a splitter, a delay applying portion, a polarization converter, a coupler, and a connecting optical waveguide; and implementing the depolarizer adjustment method according to claim **18**.
